

How to Enlarge the Bandwidth Without Increasing the Noise in OP-AMP-Based Transimpedance Amplifier

Carmine Ciofi, *Member, IEEE*, Felice Crupi, Calogero Pace, and Graziella Scandurra

Abstract—Increasing the bandwidth without degrading the noise performance represents the main challenge in the design of transimpedance amplifiers. This paper presents a novel circuit topology for a transimpedance amplifier that allows obtaining an improved tradeoff between equivalent input noise and bandwidth with respect to the conventional approach. The effectiveness of the new topology has been demonstrated by designing and testing a prototype of a transimpedance amplifier based on the proposed topology.

Index Terms—Amplifier noise, noise measurement, spectral analysis, transimpedance amplifier.

I. INTRODUCTION TO THE PROBLEM

A LARGE number of studies have proven that current noise measurements can be used as a powerful tool for the investigation of the charge transport mechanisms and of the material defectiveness in electron devices [1]–[5]. As semiconductor devices have shrunk in size, the measurement of the current noise becomes more challenging. In many cases, the performances of commercially available instrumentation are not sufficient for granting the desired sensitivity and accuracy, and therefore, the design of specialized instrumentation becomes mandatory [3]. In this paper, we propose a new circuit topology for the realization of a high-bandwidth and low-noise transimpedance amplifier that allows obtaining significant advantages with respect to more conventional designs. The most commonly used circuit topology for current noise measurement is the operational-amplifier (op-amp)-based transimpedance amplifier reported in Fig. 1. It can be easily proven that in the virtual short-circuit approximation, the frequency response is characterized by a pole at a frequency $f_R = 1/(2\pi R_R C_R)$, where C_R represents the parasitic capacitance of the feedback resistor R_R . Inasmuch as C_R is essentially independent of the resistance (for a given resistor technology), the higher the feedback resistance, the narrower will be the bandwidth. In a situation in which the bandwidth depends on the parasitic capacitance of the feedback resistor, just as it is in the case with very high resistances ($R_R > 100 \text{ M}\Omega$), there is no use in employing wide-bandwidth op-amps to extend the frequency

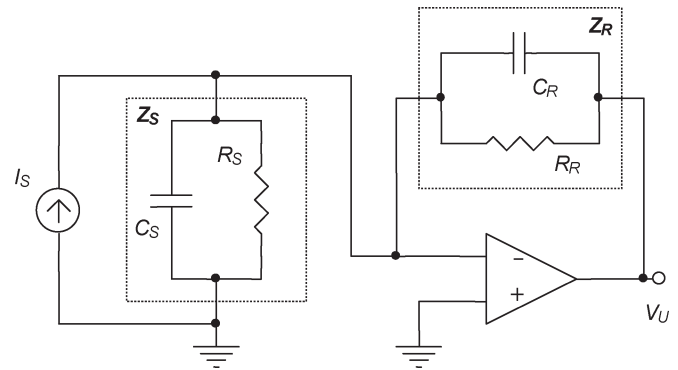


Fig. 1. Schematics of the conventional transimpedance amplifier together with the input current source with its internal impedance.

range in which the short-circuit assumption holds [6]. It can be calculated using standard analysis techniques that the power spectral density (PSD) of the equivalent input current noise is given by

$$S_{\text{ineq}} = S_{\text{in}} + \frac{S_{\text{vn}}}{|Z_S/Z_R|^2} + \frac{4kT}{R_R} \quad (1)$$

where S_{in} and S_{vn} represent the PSD of the equivalent input current and voltage sources of the op-amp. The contribution of S_{in} with respect to the other terms is usually negligible for field-effect transistor (FET) input op-amps (in the case of the TLC072, $S_{\text{in}} = 3.6 \times 10^{-31} \text{ A}^2/\text{Hz}$). Because of the capacitive component of Z_S , the contribution due to the equivalent input voltage noise source becomes significant only at higher frequencies, and often, the most important contribution to the background noise in the bandwidth of the amplifier is due to R_R . Inasmuch as the equivalent input noise level and the bandwidth are both inversely proportional to R_R , any attempt to increase the bandwidth results in an increase of the background noise level, thus making the realization of high-bandwidth low-noise transimpedance amplifiers quite challenging.

Significant improvements can be obtained by operating the amplifier at very low temperatures, thus reducing the thermal noise contribution of the feedback resistor [7], [8] or by resorting to high-sensitivity noise measurement techniques [9]–[12]. These solutions, although effective, often result in an undesirable complication of the measurement setup. As a possible alternative, we may resort to different circuit configurations capable of overcoming the limitation of the conventional approach. In the next section, after discussing possible

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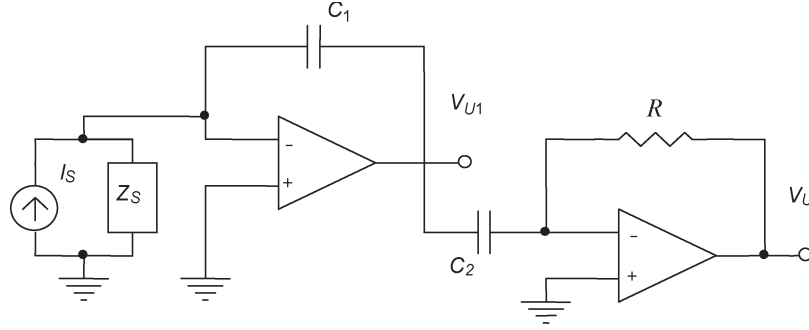


Fig. 2. Circuit schematic of the proposed low-noise high-bandwidth transimpedance amplifier.

improvements to the conventional design, we will propose a new topology that has shown to be quite effective with respect to the problem of realizing wide-bandwidth low-noise transimpedance amplifiers.

II. PROPOSED SOLUTION

An improvement to the tradeoff between noise and bandwidth can be simply obtained by **substituting the single feedback resistor with several resistors connected in series**. If we consider N equal resistors connected in series, each one with its own parasitic capacitance C , the total impedance is given by

$$Z = \frac{NR}{1 + sRC}. \quad (2)$$

This result suggests that, in principle, by increasing N , it is possible to increase the transimpedance low-frequency gain, thus reducing the corresponding thermal noise, without degrading the bandwidth. This conclusion is not completely correct because we have ignored the additional parasitic capacitance C_P that is present between the two external terminals of the resistor chain. C_P is mainly due to the capacitance between tracks and between op-amp pins. When this capacitance is also taken into account, the total impedance is given by

$$Z = \frac{NR}{1 + sR(C + NC_P)}. \quad (3)$$

This formula indicates that only when $N \ll C/C_P$, the frequency response will be characterized by a pole at a frequency $f_R = 1/(2\pi RC)$, which is independent of N . On the other hand, if $N \gg C/C_P$, the pole frequency is $f_R = 1/(2\pi NRC_P)$, which is inversely proportional to N , thus eliminating any advantage gained by increasing N . As it will be shown in the experimental section, by using this simple approach, it is difficult to gain more than a factor of 2 in the bandwidth.

To obtain superior performance in terms of bandwidth and noise, it is then mandatory to employ a different topology for realizing a transimpedance amplifier. As the **most important noise contribution in the bandwidth of the amplifier comes from the feedback resistor**, we exploited the possibility of using a noise-free dipole for Z_R , which is a purely reactive feedback impedance. The easiest choice is that of using a capacitor C_1 for Z_R . In such a case, assuming the virtual short circuit at the op-amp input, we would obtain a transimpedance gain

$A_{R1} = V_{U1}/I_S = -1/(j2\pi fC_1)$. Inasmuch as it is desirable to have a transimpedance gain that is independent of the frequency, we are then led to introduce a second stage with a voltage gain A_2 such that the overall transimpedance gain A_R is a constant value A_{R0} independent of the frequency; that is, $A_2 = V_U/V_{U1} = -j2\pi fC_1A_{R0}$. A circuit that, in principle, would allow obtaining these results is reported in Fig. 2. In the virtual short-circuit approximation, we would obtain

$$A_R(f) = \frac{V_U}{I_S} = -\frac{1}{j2\pi fC_1} \times (-j2\pi fC_2R) = \frac{C_2}{C_1}R = A_{R0}. \quad (4)$$

To compare the performance of this new circuit with respect to the conventional one, we assume the same absolute value of the low-frequency transimpedance gain, i.e.,

$$R_R = \frac{C_2}{C_1}R = A_{R0}. \quad (5)$$

For the sake of simplicity, we have neglected the parasitic capacitor in parallel with R . It must be noted, however, that if the ratio between C_2 and C_1 is large enough, R can be considerably smaller than R_R while obtaining the same overall low-frequency transimpedance gain, and therefore, the effects of its parasitic capacitance are considerably reduced. Moreover, we must note that for the circuit in Fig. 2 to work, we must provide a direct current (dc) path for biasing the inverting input of the op-amp in the first stage and for allowing to close the feedback loop at low frequencies. This can be done by connecting a resistor R_{C1} with a very large value in parallel with C_1 . It can be easily demonstrated that if we also connect a resistor R_{C2} in parallel with the capacitor C_2 in such a way as to obtain $R_{C1}C_1 = R_{C2}C_2$, the gain in the frequency range in which the short-circuit approximation holds remains unchanged and equal to that given by (5) down to dc. Moreover, using the resistors in parallel to the capacitances does lead to a finite A_{R1} at very low frequencies without changing the overall frequency response. As we shall see in the following, R_{C1} (and, accordingly, R_{C2}) should be chosen large enough in order not to add a significant contribution to the equivalent input noise. The actual circuit we used for the measurements is reported in Fig. 3. The PSD of the equivalent input current noise is given by

$$S_{ineq} = \frac{S_{vn1}}{|Z_S/Z_1|^2} + \frac{S_{vn2}}{|R_R/Z_1|^2} + \frac{C_1}{C_2} \frac{4kT}{R_R} + \frac{4kT}{R_{C1}} \left(1 + \frac{C_1}{C_2}\right) \quad (6)$$

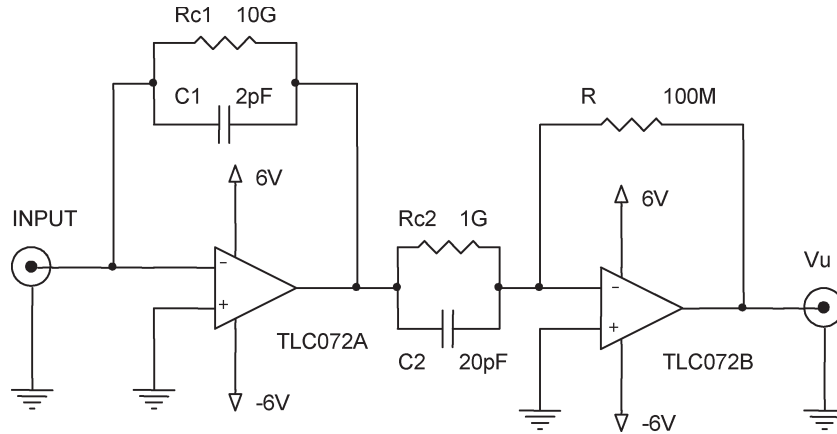


Fig. 3. Simple implementation of the proposed design for an ultra-low-noise high-bandwidth transimpedance amplifier.

where S_{vn1} and S_{vn2} represent the PSD of the equivalent voltage sources of the op-amps used in the first and second stages, respectively.

On the one hand, for the same transimpedance gain A_{R0} , the new design allows to extend the bandwidth of the amplifier with respect to the conventional design. In particular, assuming that the bandwidth is due in both cases to the effect of the parasitic capacitor in parallel with R_R (conventional design) and with R (new design), the gain in the bandwidth is proportional to the ratio C_2/C_1 . On the other hand, for the same transimpedance gain A_{R0} , and provided we can employ a resistance R_{C1} large enough with respect to R , the new design allows a reduction of the thermal noise component by a factor C_2/C_1 .

It is worth noticing that, in the above discussion, we have assumed that the poles introduced by the actual frequency response of the op-amps occur at frequencies higher than that corresponding to the pole introduced by the parasitic capacitance in parallel to R . Although this is often the case when dealing with transresistance gains higher than 100 M Ω , one may need to perform a detailed analysis or simulation of the frequency response of the entire system to verify the actual bandwidth that can be obtained for a given type of op-amp.

The drawback of the new design with respect to the conventional one is the appearance of **an additional noise component due to the equivalent input noise source V_{n2} of the op-amp used in the second stage**. Moreover, there is an increase of the contribution of the equivalent input noise source of the first stage at higher frequencies due to the presence of C_1 . In principle, we could take full advantage out of the new design by using a capacitor C_1 with a very small value while maintaining the ratio C_2/C_1 unchanged. However, there are limitations to the values that the capacitor may assume. Indeed, besides the practical difficulties in realizing capacitors with very small values using discrete components, one must take into account the fact that using very small value capacitors would lead to a reduction of the bandwidth of the amplifier because of a reduction of the loop gain of the first stage when the input impedance Z_S is taken into account. Moreover, another important limitation prevents the reduction of this noise contribution. Indeed, so far, we have neglected the presence of **common-mode capacitances at the input of the op-amps**. Such capacitances can be as large

as 20 pF (TLC2201 and TLC072), and they appear, as far as the noise contribution is concerned, in parallel to Z_S and Z_2 (common-mode capacitances between the inverting inputs and the ground of the first and second op-amps), thus posing the ultimate limit to the background noise of the system. It must be noted, however, that such an effect is of course present also in the conventional design. Moreover, in the limit in which C_1 becomes negligible with respect to the input common-mode capacitance, the additional noise contribution due to this effect would be quite limited, especially in the case in which the device-under-test (DUT) input impedance has a significant capacitive component as it is the case in most practical applications [e.g., noise measurements on metal–oxide–semiconductor (MOS) capacitors].

III. EXPERIMENTAL RESULTS

The proposed circuit solutions for large-bandwidth low-noise transimpedance amplifiers were realized **by using surface mount technology (SMT)** because very high value resistors (higher than 10 G Ω) are available in SMT. Moreover, this technology is characterized by lower parasitic capacitances and reduced sizes. The possibility of realizing extremely compact transresistance amplifiers can be quite important in the case in which one is willing to perform noise measurements at wafer level because, in such a case, the amplifier should be as close as possible to the contacting probe tip.

Fig. 4 compares the PSD of the equivalent current noise obtained by using 1- and 10-G Ω feedback resistor in a conventional design. As it was expected, by increasing the resistance, the PSD of the thermal noise contribution is ten times lower, but the bandwidth decreases by the same factor, from 60 to 6 Hz. It is worth noting that due to the availability of SMT resistors with very high resistances, we can easily realize transimpedance amplifiers with ultra-low input current noise, down to about 1 fA/ $\sqrt{\text{Hz}}$, but with a very limited bandwidth.

Initially, we investigated the improvement that could be obtained by simply connecting the feedback resistors in series. Fig. 5 shows the spectra of the equivalent current noise obtained by using a single 10-G Ω resistor or two 5-G Ω resistors connected in series. It must be noted that the series connection

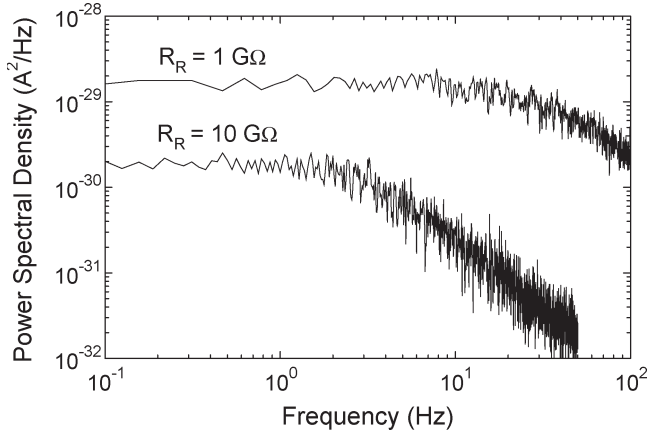


Fig. 4. PSD of the equivalent input current noise for the standard transimpedance amplifier with 1- and 10-G Ω resistor as input. The equivalent input noise level and the bandwidth are both inversely proportional to the feedback resistance.

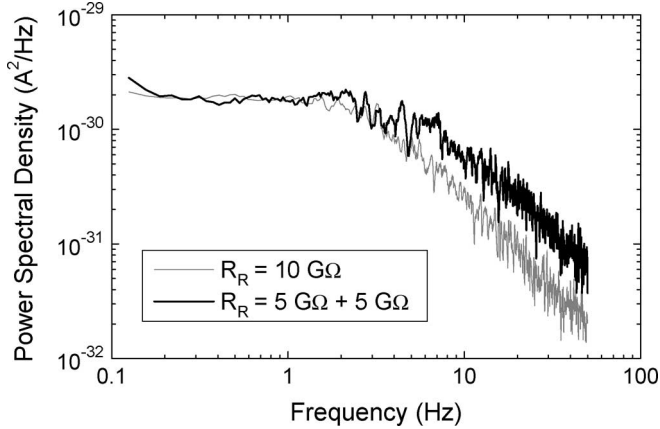


Fig. 5. PSD of the equivalent input current noise for the standard transimpedance amplifier with a single 10-G Ω resistor and two 5-G Ω resistors as input. By using two resistors connected in series, we obtain the same low-frequency noise and a larger bandwidth.

allows an increase in the bandwidth (9 Hz instead of 6 Hz) for the same input noise level. However, the observed bandwidth increase was lower than a factor of 2, as it was expected for negligible C_P .

Afterward, we performed the experiments using the new transimpedance amplifier topology shown in Fig. 3. A photograph of the realized amplifier is reported in Fig. 6. To further increase the bandwidth of the amplifier, we selected for the resistance R a TYCO HA type resistor that is characterized by a very low parasitic capacitance (on the order of 100 fF). In Fig. 7, we report the PSD of the equivalent current noise source at the input that would produce the same measured output voltage noise power. This has been done by simply dividing the output voltage PSD by the modulus of the low-frequency transimpedance gain squared. The input of the transimpedance amplifier was left open. However, because of the input common-mode capacitance C_{CM} of the op-amps, an equivalent impedance $Z_S = 1/(j2\pi f C_{CM})$, with $C_{CM} = 20$ pF, was actually present in parallel with the input port. The obtained current noise level is 1.9 fA/ $\sqrt{\text{Hz}}$ at lower frequencies. We obtain a bandwidth of about 20 kHz, as shown in

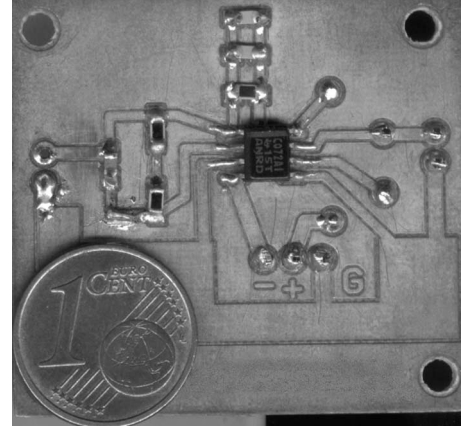


Fig. 6. Photograph of the compact circuit for ultra-low-noise current measurement. The resistor R is not visible in the picture as it is mounted from the other side of the printed circuit board (PCB).

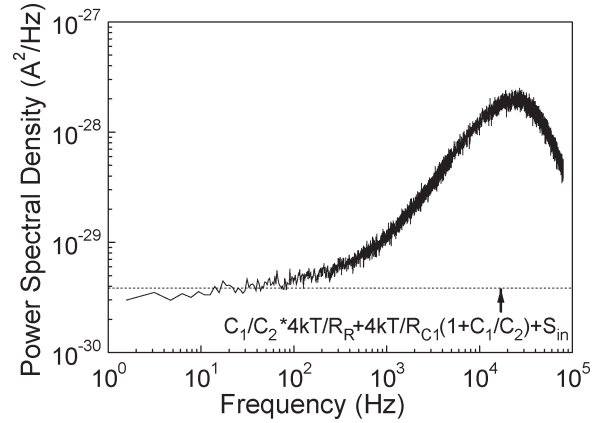


Fig. 7. PSD of the equivalent input current noise for the proposed transimpedance amplifier with the input left open. The obtained current noise level is 1.9 fA/ $\sqrt{\text{Hz}}$ at lower frequencies.

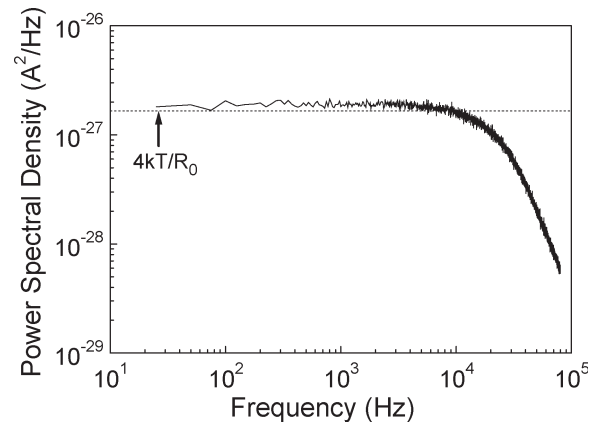


Fig. 8. PSD of the equivalent input current noise for the proposed transimpedance amplifier with $R_0 = 10$ -M Ω resistor as input. The obtained bandwidth is 20 kHz.

Fig. 8. Inasmuch as we used a C_2/C_1 ratio of 10, we expect a maximum reduction of 10 in the white noise component with respect to a conventional transimpedance amplifier for the same value of the transimpedance gain (1 G Ω). As can be simply evaluated, however, we obtained a reduction of about 5, and this was because of the contribution of the white noise introduced

TABLE I
COMPARISON OF THE PERFORMANCES OF DIFFERENT
TRANSIMPEDANCE AMPLIFIERS

Same equivalent input noise ($\approx 2 \text{ fA}/\sqrt{\text{Hz}}$)			
	New design ($1.9 \text{ fA}/\sqrt{\text{Hz}}$)	SR570 ($5 \text{ fA}/\sqrt{\text{Hz}}$)	LCA-200-10G ($1.5 \text{ fA}/\sqrt{\text{Hz}}$)
Bandwidth:	20 kHz	10 Hz	200 Hz
Same bandwidth ($\approx 20 \text{ kHz}$)			
	New design (20 kHz)	SR570 (20 kHz)	LCA-20K-200M (20 kHz)
Equivalent Input Noise:	$1.9 \text{ fA}/\sqrt{\text{Hz}}$	$600 \text{ fA}/\sqrt{\text{Hz}}$	$14 \text{ fA}/\sqrt{\text{Hz}}$

by R_{C1} . Such a resistance, as we have observed before, can be made much bigger, thus allowing obtaining the full potential advantage of the new design: No other parameter depends on R_{C1} , and increasing its value is just a matter of overcoming the usual technical difficulties that are encountered in retrieving and employing very high resistances in electronic circuits.

If we compare our new design with commercial instrumentation (the popular SR570 by Stanford Research, and two among the best amplifiers available on the market by FEMTO), we obtain better performances both when we compare the bandwidth for the same equivalent input noise and when we compare the equivalent input noise for the same bandwidth (Table I).

As it can be noted in Fig. 8, at about 1 kHz, the noise contribution due to the equivalent input noise voltage sources of the op-amps becomes significant. Reducing such noise contribution requires either resorting to op-amps with smaller equivalent input noise and/or to op-amps with smaller input common-mode capacitance (C_1 and C_2 are of the same order of C_{CM}). It must be noted that the decrease of the noise power that occurs above about 20 kHz is actually due to the fact that the frequency response of the amplifier starts to **decrease because of the combined effect of the pole introduced by the parasitic capacitance across R and the poles introduced by the op-amps that were not accounted for in the limit of the virtual short-circuit approximation.**

IV. CONCLUSION

This paper addresses the problem of increasing the bandwidth without degrading the noise performance in the design of transimpedance amplifiers. We propose a solution to this problem based on a new topology of **ultra-low-noise wide-bandwidth transimpedance amplifier**. When compared with the conventional approach for the same low-frequency transimpedance gain, the new design allows extending the bandwidth and reducing the thermal noise contribution of the feedback resistor. It is worth noting that these advantages have been obtained without significant complication in the design. The effectiveness of the new topology has been demonstrated by designing and testing a prototype of a transimpedance amplifier with the following quite remarkable characteristics: an ultra-low equivalent input current noise of $1.9 \text{ fA}/\sqrt{\text{Hz}}$ at lower frequencies and a high bandwidth of about 20 kHz.

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